

Arjun Lakshmipathy

Email: aslakshm@andrew.cmu.edu

Website: arjun-lakshmipathy.com

Github: [lakshmipathyarjun6](https://github.com/lakshmipathyarjun6)

LinkedIn: [Arjun Lakshmipathy](#)

Research Interests

Robotics: Manipulation, Soft Robotics, Motion Planning, Sensors, Locomotion

Graphics: Data Driven Animation, AR/VR, Discrete Differential Geometry, Physics-Based Simulation & Modeling

Optimization: Quadratic Programming, Convex and Non-Convex Methods, Numerical Approximation

Artificial Intelligence: Reinforcement Learning, Deep Learning, Heuristic Search, Statistical Inference

Imaging & Vision: Point Cloud Processing, Segmentation & Reconstruction Methods, Scene Understanding, Stereo Photography

Education

Carnegie Mellon University

GPA: N/A

Graduation: 2026

Ph.D. Computer Science

Advisor: Nancy Pollard

University of California, Los Angeles

GPA: 3.814

Graduation: 2018

M.S. Computer Science

Advisor: Demetri Terzopoulos

University of California, Los Angeles

GPA: 3.668

Graduation: 2016

B.S. Computer Science & Engineering

B.S. Mathematics & Economics

Research Appointments

Doctoral Research Assistant | CMU Foam Robotics Lab

September 2020 — Present

Research Scientist | NeuralX Incorporated

March 2019 — August 2020

Staff Research Associate | UCLA Computer Graphics & Vision Lab

January 2019

Graduate Student Researcher | UCLA Computer Graphics & Vision Lab

November 2017 — December 2018

Publications (Reverse Chronological)

Towards Very Low-Cost Iterative Prototyping for Fully Printable Dexterous Soft Robotic Hands

Dominik Bauer, Cornelia Bauer, **Arjun Lakshmipathy**, Roberto Shu, Nancy S. Pollard

Manuscript - Under Review at IEEE International Conference on Soft Robotics (ROBOSOFT'22)

Contact Transfer: A Direct, User-Driven Method for Human to Robot Transfer of Grasps and Manipulations

Arjun Lakshmipathy, Dominik Bauer, Cornelia Bauer, Nancy S. Pollard

Manuscript - Under Review at IEEE International Conference on Robotics and Automation (ICRA'22)

Fully Printable Low-Cost Dexterous Soft Robotic Manipulators for Agriculture

Dominik Bauer, Cornelia Bauer, **Arjun Lakshmipathy**, Nancy S. Pollard

To Appear at AAAI Conference on Artificial Intelligence - AI for Agriculture and Food Systems Workshop (AAAI AIFS'22)

Contact Tracing: A Low Cost Reconstruction Framework for Surface Contact Interpolation

Arjun Lakshmipathy, Dominik Bauer, Nancy S. Pollard

Proc. IEEE International Conference on Intelligent Robots and Systems (IROS'21), Prague, Czech Republic, September 2021

Locally-Connected, Irregular Deep Neural Networks for Biomimetic Active Vision in a Simulated Human

Masaki Nakada, Honglin Chen, **Arjun Lakshmipathy**, Demetri Terzopoulos

Proc. 25th International Conference on Pattern Recognition (ICPR'20), Milan, Italy, January 2021, 4465–4472

Deep Learning of Neuromuscular and Visuomotor Control of a Biomimetic Simulated Humanoid

Masaki Nakada, Tao Zhou, Honglin Chen, **Arjun Lakshmipathy**, Demetri Terzopoulos

IEEE Robotics and Automation Letters, 5(3), July 2020, 3952–3959. (Date of publication: February 10, 2020.)

Biomimetic Eye Modeling and Deep Neuromuscular Oculomotor Control

Masaki Nakada*, **Arjun Lakshmipathy***, Honglin Chen, Tao Zhou, Xin Ling, Demetri Terzopoulos

ACM Transactions on Graphics, 38(6), November 2019, 221:1–14. *Proc. ACM SIGGRAPH Asia 19 Conference, Brisbane, Australia, November 2019*

(* equal contributors)

Teaching Experience

Introduction to Computer Graphics (TA) | UCLA CS 174A

Fall 2017, Winter 2018

Basic principles behind modern two- and three-dimensional computer graphics systems, including complete set of steps that modern graphics pipelines use to create realistic images in real time. How to position and manipulate objects in scene using geometric and camera transformations. How to create final image using perspective and orthographic transformations. Basics of modeling primitives such as polygonal models and implicit and parametric surfaces. Basic ideas behind color spaces, illumination models, shading, and texture mapping.

Computer Networks: The Physical Layer (TA) | UCLA CS 117

Spring 2018, Summer 2018, Fall 2018

Introduction to fundamental computer communication concepts underlying and supporting modern networks, with focus on wireless communications and media access layers of network protocol stack. Systems include wireless LANs (IEEE802.11) and ad hoc wireless and personal area networks (e.g., Bluetooth, ZigBee). Experimental project based on mobile radio-equipped devices (smart phones, tablets, etc.) as sensor platforms for personal applications such as wireless health, positioning, and environment awareness, and experimental laboratory sessions included.

Technical Skills

C++ (Dynamic Animation and Robotics Toolkit, Point Cloud Library, OpenSceneGraph, Eigen, ImGUI, OpenGL + GLUT), Python (NumPy, Open3D, Tensorflow, PyTorch, SciPy), C, ROS, Blender, AWS (EC2, S3, EKS, ELB, ECS, CloudFront, Cognito, DynamoDB, MediaConvert, Route 53, ACM, SES, SNS), JavaScript (NodeJS / Express, ReactJS, AngularJS), Docker + Kubernetes, NGINX + Certbot, Jenkins, Java / Spring, Scala / Play, Elasticsearch, Matlab, Git

Graduate Coursework

Graduate Algorithms (CMU 15-750)	<i>Spring 2021</i>
Biomechanics and Motor Control (CMU 16-868)	<i>Fall 2020</i>
Computational Photography (CMU 15-862)	<i>Fall 2020</i>
Problem Solving and Search (UCLA CS 261A)	<i>Spring 2018</i>
Special Topics: Shared Economy Systems (UCLA CS 259)	<i>Spring 2018</i>
Artificial Life for Computer Graphics and Vision (UCLA CS 275)	<i>Winter 2018</i>
Neural Networks and Deep Learning (UCLA EE 239AS)	<i>Winter 2018</i>
Embedded Systems (UCLA CS M213A)	<i>Fall 2017</i>

Non-Research Professional Appointments*

Senior Tutor (Part Time)

March 2021 — Present

Cambridge Coaching

Technical Advisor (Part Time)

September 2020 — December 2020

NeuralX Incorporated

Founding Engineer

March 2019 — August 2020

NeuralX Incorporated

Full Stack Software Engineer

January 2017 — September 2017

Rally Health Incorporated

Software Engineering Intern
Yahoo! Incorporated

June 2015 — September 2015

Software Engineering Intern
ViaSat Incorporated

June 2014 — September 2014

iOS Development / Quality Control Intern
Sandia National Laboratories

July 2013 — September 2013

(*job titles only - full description and contributions available on LinkedIn)

Service

Reviewer, IEEE International Conference on Intelligent Robots and Systems (IROS) 2021

2021

UPE + TBP Free Tutoring

2013 — 2016

Honors and Awards

Cum Laude Honors (BS)

June 2016

Upsilon Pi Epsilon Honor Society Member

April 2013 — Present

Tau Beta Pi Honor Society Member

April 2013 — Present

References

1. **Nancy Pollard**

Professor
Computer Science Department, Robotics Institute
Carnegie Mellon University
npollard@andrew.cmu.edu

2. **Demetri Terzopoulos**

Distinguished Professor and Chancellor's Professor of Computer Science
Computer Science Department
University of California, Los Angeles
dt@cs.ucla.edu

3. **Masaki Nakada**

Co-Founder and Co-CEO
NeuralX Incorporated
masaki@neuralx.ai